

Seeding Markov Chain Monte Carlo algorithms

Gutiérrez-Peña, Eduardo
eduardo@sigma.iimas.unam.mx

Aguirre-Pérez, Román
roman@sigma.iimas.unam.mx

July 11, 2026

Abstract

The purpose of this work is to illustrate the need of seeding Markov Chain Monte Carlo algorithms when dealing with unobserved variables, as well as to propose a seeding procedure based on a distance-based approach for such instances.

1 Introduction

In real-world scenarios, we often aim to explain a set of traits of a phenomenon of interest under specific conditions. From a statistical perspective, we usually propose a stochastic mechanism to approximate the underlying law that is governing such a phenomenon of interest, say $\pi(\mathbf{z}|\boldsymbol{\theta})$, where $\mathbf{z}$ is a set of attributes and $\boldsymbol{\theta}$ is a random vector that fully defines the proposed stochastic mechanism. Within the Bayesian approach (Gelman et al., 2014, see), we further elicitate a probabilistic distribution for $\boldsymbol{\theta}$, say $\pi(\boldsymbol{\theta})$.

We... Section 2. We then address this issue in Section 3 where we describe the proposed algorithm for seeding, whereas in Section 4 we illustrate the situation with a case study using the R packages **NIMBLE** (de Valpine et al., 2026) and **cmdstanr** (Gabry et al., 2025). Finally, in Section 5 we conclude with....

2 MCMC in practice

Markov Chain Monte Carlo (MCMC) methods (Brooks et al., 2011) are widely used in practice for reconstructing the posterior distribution $\pi(\boldsymbol{\theta}|\mathbf{z})$. (Here we explain how they are implemented).

Bayesian Inference Using Gibbs Sampling (BUGS) (Lunn et al., 2012) and Just Another Gibbs Sampler (JAGS) (Plummer, 2003).

Hamiltonian Monte Carlo (HMC) (Neal, 2011) is the workhorse of the Stan software (Stan Development Team, 2026). On the other hand, the NIMBLE system de Valpine et al. (2017).

3 Seeding MCMC algorithms

A more challenging scenario is when we only observed a subset of $\mathbf{z}$, that is, if $\mathbf{z} = (\mathbf{x}, \mathbf{y})$ we only have measurements for $\mathbf{y}$, say $\mathbf{y}^{obs}$. In this case, we instead seek to reconstruct $\pi(\mathbf{x}, \boldsymbol{\theta}|\mathbf{y}^{obs})$, and thus, we now also have to initialise $\mathbf{x}$. Leveraging a distance-based approach (Maa et al., 1996) in tandem with the prior predictive distribution, we proposed the following seeding process for Monte Carlo algorithms when dealing with unobserved variables.

$d : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$ is a pseudo-distance. Algorithm 1 summaries such a seeding process.

Algorithm 1 Distance-based Seeding

Require: $\boldsymbol{\theta}^{(t)}$

Ensure: $\boldsymbol{\theta}^{(t)}, \mathbf{x}^{(t)}, \mathbf{y}^{(t)}$

- 1: $t = 0$
 - 2: **repeat**
 - 3: $t \leftarrow t + 1$
 - 4: $\boldsymbol{\theta}^{(t)} \sim \pi(\boldsymbol{\theta})$
 - 5: $\mathbf{x}^{(t)}, \mathbf{y}^{(t)} \sim \pi(\mathbf{x}, \mathbf{y}|\boldsymbol{\theta})$
 - 6: **until** $d(\mathbf{y}^{(t)}, \mathbf{y}^{obs}) < d^{max}$
-

4 Case study: Population with exponential growth

[Auger-Méthé et al. \(2021\)](#) provides an overview of Bayesian methods in Ecology, whereas [Monnahan et al. \(2017\)](#) propose to use HMC for making inference.

For a discrete-time setting, let us assume that a population experience an exponential growth with time-varying rate r_t , that is, $N_{t+1} = e^{r_t} N_t$ for $t = 1, 2, \dots, T - 1$. We can rewrite the time-varying rate as $r_t = f_t - m_t$, where $f_t, m_t \in \mathbb{R}^+$ are the fertility and mortality rates at time t , respectively. Now, at each time t , we can think of abundance before and after mortality, say N_t^{pre} and N_t^{post} , respectively. Thus, if M_t denotes the total mortality we have that $N_t^{post} = N_t^{pre} - M_t$.

Given the aforementioned population demographics, we can propose the following stochastic population mechanism:

$$\begin{aligned} M_t &\sim \text{Bin}(N_t^{pre}, 1 - e^{-m_t}) \\ N_{t+1}^{pre} &\sim \text{Poi}(e^{f_t} N_t^{post}) \end{aligned}$$

Furthermore,

$$\begin{aligned} \log(N_1^{pre}) &\sim \text{Poi}(e^{LN_0}) \\ \log(a_t) &\sim \text{N}(\mu_{la}, \sigma_{la}) \\ \log(b_t) &\sim \text{N}(\mu_{lb}, \sigma_{lb}) \\ \log(f_t) &\sim \text{N}(\mu_{lf}, \sigma_{lf}) \end{aligned}$$

with

$$\begin{aligned} \mu_{lm} &\sim \text{N}(m_{lm}, sd_{lm}) \\ \sigma_{lm} &\sim \text{N}(0, sd_{lm}) \\ \mu_{lf} &\sim \text{N}(m_{lf}, sd_{lf}) \\ \sigma_{lf} &\sim \text{N}(0, sd_{lf}) \end{aligned}$$

The R project ([R Core Team, 2026](#)).

5 Concluding Remarks

References

- Auger-Méthé, M., Newman, K., Cole, D., Empacher, F., Gryba, R., King, A. A., Leos-Barajas, V., Mills Flemming, J., Nielsen, A., Petris, G., and Thomas, L. (2021). A guide to state-space modeling of ecological time series. *Ecological Monographs*, 91(4):e01470.
- Brooks, S., Gelman, A., Jones, G. L., and Meng, X.-L., editors (2011). *Handbook of Markov Chain Monte Carlo*. Chapman and Hall/CRC Press, 1st edition.
- de Valpine, P., Paciorek, C., Turek, D., Michaud, N., Anderson-Bergman, C., Obermeyer, F., Wehrhahn Cortes, C., Rodríguez, A., Temple Lang, D., and Paganin, S. (2026). *NIMBLE: MCMC, Particle Filtering, and Programmable Hierarchical Modeling*. R package version 1.4.2.
- de Valpine, P., Turek, D., Paciorek, C., Anderson-Bergman, C., Temple Lang, D., and Bodik, R. (2017). Programming with models: writing statistical algorithms for general model structures with NIMBLE. *Journal of Computational and Graphical Statistics*, 26:403–413.
- Gabry, J., Češnovar, R., Johnson, A., and Bronder, S. (2025). *cmdstanr: R Interface to 'CmdStan'*. R package version 0.9.0.
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2014). *Bayesian Data Analysis*. Chapman and Hall/CRC Press, 3rd edition.
- Lunn, D., Jackson, C., Best, N., Thomas, A., and Spiegelhalter, D. (2012). *The BUGS Book: A Practical Introduction to Bayesian Analysis*. Chapman and Hall/CRC Press, 1st edition.
- Maa, J.-F., Pearl, D. K., and Bartoszyński, R. (1996). Reducing multi-dimensional two-sample data to one-dimensional interpoint comparisons. *The Annals of Statistics*, 24(3):1069–1074.
- Monnahan, C. C., Thorson, J. T., and Branch, T. A. (2017). Faster estimation of bayesian models in ecology using hamiltonian monte carlo. *Methods in Ecology and Evolution*, 8(3):339–348.

- Neal, R. M. (2011). MCMC using Hamiltonian dynamics. In Brooks, S., Gelman, A., Jones, G. L., and Meng, X.-L., editors, *Handbook of Markov Chain Monte Carlo*, pages 113–162. CRC Press.
- Plummer, M. (2003). JAGS: A program for analysis of bayesian graphical models using gibbs sampling. In *Proceedings of the 3rd International Workshop on Distributed Statistical Computing (DSC 2003)*, Vienna, Austria.
- R Core Team (2026). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.
- Stan Development Team (2026). *Stan User’s Guide*. Version 2.39.0.